

Atonu Ghosh

Ph.D. Scholar (Thesis Submitted)

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Professional Summary

- Submitted Ph.D. thesis in *Computer Science & Engineering* at *IIT Kharagpur, India* and finished a Research Internship at *Samsung R&D*, Bangalore, India.
- 6+ years of hands-on experience in building secure, scalable IoT architectures, focusing on Edge Computing, LPWANs, and Secure Cyber-Physical Systems.
- Filed 16 patents (3 granted, 4 commercialized) and authored 12 papers (8 journal articles & 4 conference papers) across top-tier venues, including IEEE TII, TNSE, TICPS, IoT Journal, and others.
- Research focuses on *secure and scalable IoT architectures* such as *TLS-over-LoRa* and *edge system integration with AI/ML* for industrial condition monitoring.
- Led end-to-end system prototyping and field validation through consulting and research projects with *Tata Steel Downstream Products Ltd., India*, *CPHEEO*, and *MeitY*, Government of India.
- Dedicated to advancing next-generation *IoT*, *Edge Computing*, and *Low Power Network* research, translating these innovations into scalable, human-centric technologies that create tangible societal impact.

Career Highlights

Patents (Granted)	3	Patents (Pending)	13
Patents Commercialized / In use	4	Journal Articles (Published)	5
Journal Preprints / Communicated	3	Conference Articles	4

Education

2021–2026: **Ph.D., Computer Science & Engineering.**

Indian Institute of Technology (IIT), Kharagpur, India

Supervisor: Prof. Sudip Misra

Thesis Title: “*Architectures and Protocols for Reliable and Secure Low Power IoT*”

IoT, IIoT, Edge Computing, Low Power Wide Area Networks (LPWANs), Condition Monitoring

2018–2020: **Master of Technology, Computer Science & Engineering.**

Maulana Abul Kalam Azad University of Technology (formerly WBUT), Nadia, West Bengal, India

CGPA: 8.93/10

Project on design and implementation of logging system using Ethereum blockchain and IPFS

2014–2017: **Bachelor of Technology, Computer Science & Engineering.**

Institute of Engineering and Management (IEM), Salt Lake, Kolkata, India

CGPA: 7.71/10

Project on implementation of a simple operating system accepting commands

Patents

Granted

- PG3 **Atonu Ghosh**, Biswajit Ghosh, Ruelia Saha, Sudip Misra, and Arijit Roy, System for Real Time Intrusion Detection, Actuation and Alert and Method Thereof, *Indian Patent*, Number – 566326, May 16, 2025, (**Commercialized**).

- PG2 Subhas Chandra Misra, Debanjan Das, Sudip Misra, Venkanna Udutalapally, **Atonu Ghosh**, and Tanushree Pan, Wireless Network Device for Wireless Communication with User Devices in a Wireless Communication Network, *Indian Patent*, Number – 539308, May 27, 2024.
- PG1 Subhas Chandra Misra, Debanjan Das, Venkanna Udutalapally, Sudip Misra, and **Atonu Ghosh**, Blockchain-Enabled IoT System and Method for Securing Real Time Data In a Microcontroller-Based Blockchain Network, *Indian Patent*, Number – 529261, Mar 20, 2024.
[Pending](#)
- PF13 Syam Sidhardhan, **Atonu Ghosh**, Hub-Side Intelligence for Vendor Agnostic IoT Sensor Battery Life Extension, *Indian Patent*, File No. – 202641080289, Jun 30, 2026.
- PF12 Sudip Misra, **Atonu Ghosh**, Spandan Duari, Kumarjit Ray, Rajdeep Ghosh, Panagiotis Sarigiannidis, Nikos Ntampakis, and Vasileios Argyriou, System and Method for Automated Sensor Identification and Data Delivery Over Universal Serial Bus, *Indian Patent*, File No. – 202531114519, Nov 20, 2025.
- PF11 Sudip Misra, **Atonu Ghosh**, Bontalakoti Aditya, Kumarjit Ray, and Rajdeep Ghosh, An Intelligent Liquid-Level Sensing and Pump Control System, *Indian Patent*, File No. – 202531098034, Oct 10, 2025, (**Commercialized**).
- PF10 Sudip Misra, **Atonu Ghosh**, Biswajit Ghosh, Swaraj Lahiri, Bharat Bhushan, Subrata Ray, and Sonali Pan, Embedded System and Method for Condition Monitoring of Electric Overhead Travelling (EOT) Cranes, *Indian Patent*, File No. – 202531033456, Apr 4, 2025 (**Deployed at Tata Steel Downstream Products Ltd., Jamshedpur, India**).
- PF9 Aparesh Mahapatra, Biswajit Ghosh, **Atonu Ghosh** Suryansh Sharma, Debayan Biswas, Sudip Misra, Partha Sarathi Ghoshal, and Manoj Kumar Tiwari, IoT-Enabled Smart Hardware System with Edge Intelligence for Rapid Leak Detection in Water Distribution Networks, *Indian Patent*, File No. – 202531028121, Mar 25, 2025 (**Deployed for Water Distribution Pipe Monitoring at Haldia, WB, India**).
- PF8 **Atonu Ghosh** and Sudip Misra, LoRaNext: Embedded System and Method for Decentralized HTTP Communication Over LoRa, *Indian Patent*, File No. – 202531024600, Mar 19, 2025.
- PF7 **Atonu Ghosh**, Sudip Misra, and Sharath Chandan, Long Range Multiple Input Multiple Output System for High Throughput Communication, *Indian Patent*, File No. – 202431094880, Dec 2, 2024.
- PF6 Kumarjit Ray, **Atonu Ghosh**, Biswajit Ghosh, and Sudip Misra, Universal Software-Defined LoRa Gateway Based Network System for Remote Programming and Management of Network Functions, *Indian Patent*, File No. – 202431046371, Jun 15, 2024.
- PF5 Ruelia Saha, **Atonu Ghosh**, and Sudip Misra, Confidant: An Internet of Things-Based System and Method for Mental Health Monitoring and Support, *Indian Patent*, File No. – 202431034007, Apr 29, 2024.
- PF4 Biswajit Ghosh, **Atonu Ghosh**, and Sudip Misra, An Edge Intelligence Based Programmable Logic Controller, *Indian Patent*, File No. – 202431032957, Apr 25, 2024.
- PF3 **Atonu Ghosh**, Sudip Misra, Arijit Roy, and Anandarup Mukherjee, A System for Remote Monitoring, Actuation and Data Prediction, *Indian Patent*, File No. – 202231040824, Jul 17, 2022.
- PF2 Sudip Misra, Saswati Pal, and **Atonu Ghosh**, Modularized IoT-Based On-Demand Ambulatory Hospital Recommender System, *Indian Patent*, File No. – 202231008007, Feb 15, 2022.
- PF1 Sudip Misra, Debanjan Das, Venkanna Udutalapally, **Atonu Ghosh**, and Pallav Kumar Deb, A Secured Edge-Based Automated Power Control and Communication System for Legacy IoT Infrastructures, *Indian Patent*, File No. – 202131038016, Aug 23, 2021.

Publications

Journal Articles

- J5 **Atonu Ghosh**, Akhilesh Mohanasundaram, Srishivanth RF, and Sudip Misra. *Siaa: Secure Industrial IoT Architecture for HTTPS Communication Over LoRa Using TLS*. *IEEE Transactions on Industrial Informatics*, 2026, DOI: <https://doi.org/10.1109/TII.2026.3713181>.
- J4 **Atonu Ghosh**, Sudip Misra, Venkanna Udutalapally, and Debanjan Das. *LoRaute: Routing Messages in Backhaul LoRa Networks for Underserved Regions*. *IEEE Internet of Things Journal*, volume 10, pages 19964–19971, 2023, DOI: <https://doi.org/10.1109/JIOT.2023.3281941>.
- J3 **Atonu Ghosh**, Sudip Misra, and Venkanna Udutalapally. *Multiobjective Optimization and Sensor Correlation Framework for IoT Data Validation*. *IEEE Sensors Journal*, volume 22, pages 23581–23589, 2022, DOI: <https://doi.org/10.1109/JSEN.2022.3215993>.
- J2 **Atonu Ghosh**, Ruelia Saha, and Sudip Misra. *Persistent Service Provisioning Framework for IoMT Based Emergency Mobile Healthcare Units*. *IEEE Journal of Biomedical and Health Informatics*, volume 26, pages 5851–5858, 2022, DOI: <https://doi.org/10.1109/JBHI.2022.3172624>.
- J1 **Atonu Ghosh**, Anandarup Mukherjee, and Sudip Misra. *SEGA: Secured Edge Gateway Microservices Architecture for IIoT-Based Machine Monitoring*. *IEEE Transactions on Industrial Informatics*, volume 18, pages 1949–1956, 2021, DOI: <https://doi.org/10.1109/TII.2021.3102158>.

Communicated Journal Articles and Preprints

- JC3 Apareesh Mahapatra, **Atonu Ghosh**, Sudip Misra, Sharath Chandan, and Partha Sarathi Ghosal, *Breaking the Throughput Barrier of LoRa in IoT: A Multi-Radio MIMO System for WDNs*, in *IEEE Internet of Things Journal*, (In Revision, 2026).
arXiv Preprint: <https://arxiv.org/abs/2501.07148v1>
- JC2 **Atonu Ghosh**, David Breuss, Simon Howind, Sudip Misra, Thilo Sauter, and Surjya Kanta Pal, *SenseAI: Staged Model Deployment for Resource Constrained AIIoT-Enabled Industrial Environments*, in *IEEE Transactions on Industrial Cyber-Physical Systems*, (In Revision, 2025).
- JC1 **Atonu Ghosh** and Sudip Misra, Shri: *Simple HTTP Resource Interchange Over LoRa Backbone Networks*, in *IEEE Transactions on Network Science and Engineering*, (In Revision, 2025).
arXiv Preprint: <https://arxiv.org/abs/2501.02469>

In Conference Proceedings

- C4 Syam Sidhardhan, T K Ramesh, **Atonu Ghosh**, Inpyo Kang, and Dongwook Lee. *MultiHub: Smart home iot hub coordination framework for energy-efficient sensing*. In *International Conference on Electronics, Computing and Communication Technologies (CONECCT)*. IEEE, 2026 (**Accepted**).
- C3 **Atonu Ghosh**, Debashis De, and Koushik Majumder. *A systematic review of log-based cloud forensics*. In *Inventive Computation and Information Technologies (ICIT)*, pages 333–347. Springer, 2021, DOI: https://doi.org/10.1007/978-981-33-4305-4_26.
- C2 **Atonu Ghosh**, Koushik Majumder, and Debashis De. *Android forensics using sleuth kit autopsy*. In *International Conference on Mathematics and Computing (ICMC)*, pages 297–308. Springer, 2020, DOI: https://doi.org/10.1007/978-981-15-8061-1_24.
- C1 **Atonu Ghosh**. *Intelligent appliances controller using raspberry pi*. In *Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, IEEE, pages 1–5. IEEE, 2016, DOI: <https://doi.org/10.1109/IEMCON.2016.7746253>.

Book Chapter

- B1 **Atonu Ghosh**, Koushik Majumder, and Debashis De, A Systematic Review of Digital, Cloud and IoT Forensics, In *The “Essence” of Network Security: An End-to-End Panorama*, Springer, 2021. DOI: https://doi.org/10.1007/978-981-15-9317-8_2.

Work Experiences

- Jan 2026 – **Research Intern, Samsung R&D Institute, Bangalore, India.**
July 2026 Worked full-time onsite on cutting-edge research and development projects on energy saving in the *SmartThings* Team.
- Jan 2021 – **Ph.D. Research Scholar, Indian Institute of Technology (IIT), Kharagpur.**
Present Under the Supervision of **Prof. Sudip Misra** on IoT, Edge Computing, Low Power Networks (LPWANs), and Condition Monitoring
- **EOT Crane Condition Monitoring at Tata Steel Downstream Products Limited**
 - **Water Distribution Pipe Pressure Monitoring at Haldia Water Services Pvt. Ltd., WB, India**
 - **IoT Enablement of Legacy Sensors at Bokaro Steel Plant**
 - **Automate Ductile Iron Pipe Coating Thickness Measurement at Tata Steel**
 - **Full TLS 1.3 Handshake Over LoRa for Secure IoT**
 - **LoRa MIMO Testbed**
 - **Implemented HTTP Over LoRa**
 - **Lightweight Blockchain on Microcontroller**
 - **Live Ambulatory Patient Monitoring System**
- Oct 2020 – **Research Fellowship, Indian Institute of Technology (IIT), Kharagpur.**
Dec 2023 **Senior Research Fellowship** (Oct 27, 2022 – Dec 31, 2023)
Junior Research Fellowship (Oct 22, 2020 – Oct 22, 2022)
- Worked under Prof. Sudip Misra in the **Ministry of Electronics and Information Technology (MeitY), Government of India**, sponsored research project entitled *TribeConnect: Integrated Smart Tribal Eco-Platform – A Proof of Concept in Chhattisgarh*.
- Project *TribeConnect* aimed at the upliftment and empowerment of the tribal community through the application of IoT and Machine Learning in the fields of agriculture and healthcare.
- **LoRa Network for Research Project Funded by MeitY, Government of India**
 - **IoT Gateway with Optional Electrical Load Controller**
 - **Live Server Management**

- **Developed IoT Dashboard & Automatic Database Replication**

Nov 2020 – **System Architect, *SensorDrops Networks Pvt. Ltd., Kharagpur.***

Oct 2025 Contributed voluntarily without pay to this startup founded by my Ph.D. supervisor.

- **Pet Immune Human Intruder Detection and Alert System**
- **Machine and Ambient Condition Monitoring System**
- **Smartphone as a Controller for Electrical Lights and Fans**

Academic Achievements & Recognitions

2024 **Qualcomm Innovation Fellowship**, Finalist

2026 Invited as **Session Chair** for Demos at the **IEEE CONECCT 2026 Conference**, Bangalore, India

Instructor in Live/Hands-on Sessions

- July, 2024 Security for Internet of Things and DevSecOps, **Tata NeuSkills**
- April, 2024 Security for Internet of Things and DevSecOps, **Tata NeuSkills**
- Jan, 2024 Short Term Course on **Hands-on** Introduction to Internet of Things with Machine Learning, **IIT Kharagpur**
- Dec, 2023 Short Term Course on **Hands-on** Introduction to Internet of Things with Machine Learning, **IIT Kharagpur**
- Dec, 2023 Security for Internet of Things and DevSecOps, **Tata NeuSkills**
- Dec, 2023 Introduction to Cloud Computing, **IIT Kharagpur**
- Jul, 2023 Short Term Course on **Hands-on** Introduction to Internet of Things with Machine Learning, **IIT Kharagpur**
- April, 2023 Introduction to Cloud Computing, **IIT Kharagpur**

Skills

- IoT Hardware Raspberry Pi, Microcontrollers (ESP32, ESP8266, Raspberry Pi Pico, 8051), LoRa Transceivers, Sensors, Actuators, Discrete Components
- Programming Languages Python, Micropython, C
- Web, Edge, & Cloud Machine Learning (ML) Integration on Edge and Cloud, Django, Docker, Kubernetes, Nginx, REST API, Swagger, Apache, HTML, CSS, JavaScript, PHP, AWS, Google Cloud, Azure, OpenStack, and Digital Ocean
- Database SQL, MySQL, PostgreSQL, MariaDB, InfluxDB, SQLite, Redis
- Tools Linux Kernel Modification and Cross Compilation, GitHub, Wireshark, Altium Designer, Fusion 360, Proteus, and Keil μ Vision, NodeRed
- Miscellaneous Mentoring Interns, Ph.D., M.S., M.Tech., and B.Tech. Students, Project & Lab Management, Academic Research, Consulting Projects, Project Proposal Writing, Selecting and Procuring Instruments and Components for Lab.

Volunteering

2022-2023 **Secretary of IEEE Computer Society Student Branch Chapter**, IIT Kharagpur.